Important Questions (Classical Physics for Engineers)22PY211IB

Sl.No.	Questions
1	What is damped oscillation? Set up the differential equation for damped harmonic oscillation and find the solution for displacement. Discuss the case of heavy damping, critical damping and under damping
2	Obtain the differential equation for the system executing angular simple harmonic oscillation and find the period of the oscillation.
3	Arrive at an expression for effective spring constant for springs in series and parallel.
4	What is forced harmonic oscillation? Set up the differential equation for forced harmonic oscillation, compare with the mechanical forced harmonic oscillation and write the equation for amplitude and phase.
5	What is resonance? Explain the resonance condition for LCR circuit.
6	Explain the behaviour of elastic material using Stress strain graph. Explain each term on stress strain graph.
7	Show that a simple shear is equivalent to the sum of equal linear compression strain and elongation strain perpendicular to each other.
8	Show that shear stress is equal to the linear tensile stress and compression stress.
9	Find the limiting value of Poisson ratio using elastic constant relation.
10	What is torsional rigidity? Derive an expression for couple per unit twist of a solid cylinder
11	What is bending moment? Derive an expression for bending moment of a beam having rectangular are of cross section.
12	Using bending moment expression, derive an expression for Young's modulus of a single cantilever.
13	With neat diagram explain Joule's experiment.
14	With neat diagram explain construction and working of liquid, gas and resistance thermometers.
15	Derive adiabatic equation of state, derive an expression for work done in adiabatic process.
16	Explain isothermal, isobaric and isochoric process.
17	What are Newtonian and non Newtonian fluids? Give example each. Explain Newton's law of viscosity.
18	Write a note on No- slip condition and cavitations.
19	With neat diagram explain the principle, construction and working of Ultrasonic interferometer.
20	With neat diagram, explain the construction and working of rotational viscometer.
21	What is surface tension of fluid? What is capillarity? Derive an expression for excess pressure inside a soap bubble and an air bubble.
22	Applying continuity equation in integral form, Set up continuity equation for a steady state fluid in pipe.
23	Define laminar flow, turbulence flow and transitional flow of a fluid. Explain Renold number
24	Define steam line, path line, streak line and timeline flow of fluids.
25	With neat diagram explain the construction and working of Universal Testing Machine (UTM).
26	Explain the determination of Hardness of the material using Rockwell Hardness testing, Brinell Hardness testing and Vickers Hardness testing technique.
27	Write a note on Magnetic particle testing for surface characterization.
28	With neat diagram explain Ellipsometry technique to determine the thickness of the film.
29	With neat graph explain IV characteristics of a solar PV cell
30	What is Bragg's law, derive Bragg's law.
31	Explain the analysis of crystal structure using Bragg's diffractometer (XRD)

32	With neat diagram explain X-ray Photoelectron Spectroscopy.
33	With neat labelled diagram, explain the construction and working of Scanning Electron Microscope (SEM) and Transmission Electron Microscope (TEM).
34	With neat labelled diagram explain construction and working of Atomic Force Microscope (AFM)